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AI Inference on AMD Ryzen™ AI Max Processor



May 25, 2026 by [Chengjia Huang](#), [Daniel Huang](#), [George Wang](#). 7 min read. | 1614 total words.

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Local large language model (LLM) inference has rapidly evolved, but a persistent limitation remains: model size is constrained by available GPU memory. Discrete GPUs typically offer 8–24 GB of dedicated VRAM, which can limit the size of models that can run without incurring significant quality loss from aggressive quantization. As frontier open-weight models grow past 70B and 100B parameters, this gap is forcing more developers toward multi-GPU rigs or paid cloud endpoints just to evaluate a single checkpoint.

AMD Ryzen™ AI Max+ processors with AMD Radeon™ 8060S integrated graphics change that equation. With up to 128 GB of unified memory shared between CPU and GPU, the entire memory pool is accessible for AI workloads — enabling models with 100 billion or more parameters to run on a single system, with no second card and no cloud bill.

If you build, evaluate, or self-host LLMs on AMD hardware, this blog gives you a concrete, reproducible path to take advantage of that. By the end you will know how to:

- Set up Ollama on Ubuntu with AMD ROCm™ technology on an AMD Ryzen™ AI Max+ processor-based system;
- Configure and reason about Unified Memory Architecture (UMA), including the GPU-accessible memory slice;
- Run Qwen3.5 across three size tiers — 9B (dense), 35B-A3B (MoE), and 122B-A10B (MoE) — and interpret Ollama’s `--verbose` performance output;
- Decide which model size best fits your latency, quality, and capacity targets on a single Ryzen™ AI Max+ processor-based system.

The **122B tier** is the headline *capacity* story (weights + runtime on one machine); all generation benchmarks shown later were collected with **Ollama** on the same AMD Ryzen™ AI Max+ processor-based system.

System Requirements

Component	Specification
Processor	AMD Ryzen™ AI Max+ 395
Integrated GPU	AMD Radeon™ 8060S Graphics (RDNA 3.5)
Total System Memory	128 GB LPDDR5X (Unified)
GPU-Accessible memory	64 GB (configurable)
Operating System	Ubuntu 24.04 LTS
AMD ROCm™ Version	7.2.1 (HSA Runtime 1.18)
Ollama Version	0.20.x

Understanding Unified Memory Architecture

Traditional discrete GPUs have dedicated VRAM separate from system RAM. A GPU with 24 GB of VRAM can only hold models that fit within that 24 GB, regardless of how much system RAM is available.

AMD Ryzen™ AI Max+ processors use a Unified Memory Architecture (UMA), where the CPU and GPU share the same physical memory pool. On a 128 GB system, the GPU-accessible portion can be configured to 64 GB or more through BIOS settings, with the remainder available for the operating system and applications.

This means models that would require multiple discrete GPUs — or cloud-hosted instances — can often run on a single AMD Ryzen™ AI Max+ processor-based system; smaller models can use 100% GPU offloading, while the largest models use CPU/GPU mixed loading when they exceed the GPU-accessible slice.

Figure 1 below contrasts unified memory with a typical discrete-VRAM model.

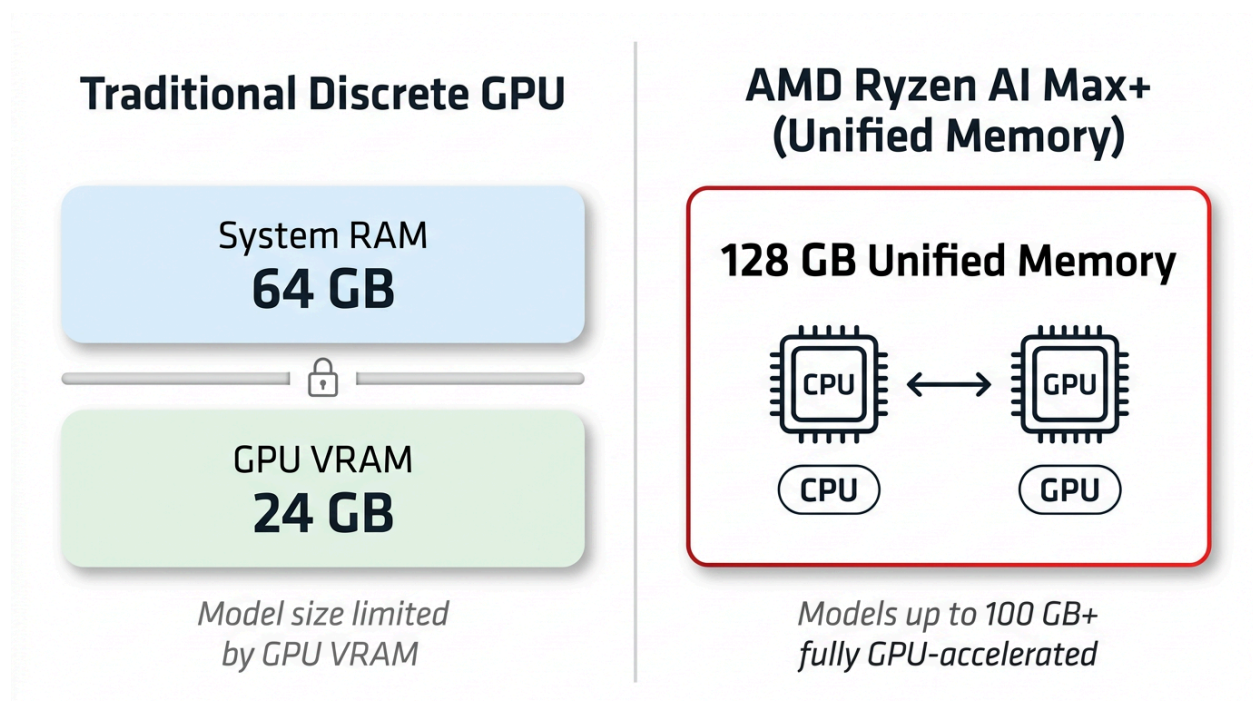


Figure 1: Unified Memory Architecture — CPU and GPU share the same 128 GB memory pool, enabling large model inference without dedicated VRAM limitations.

Installation Guide

Recommended Method: Ollama

Ollama provides a streamlined installation and model management experience. On Ubuntu 24.04 with AMD ROCm™ software already installed:

```
curl -fsSL https://ollama.com/install.sh | sh
```

Download and run the model:

```
ollama pull qwen3.5:35b  
ollama run qwen3.5:35b
```

Verify GPU acceleration is active:

```
ollama ps
```

The output should show **100% GPU**, confirming all model layers are offloaded to the AMD Radeon™ 8060S integrated GPU.

Performance Benchmarks

We tested three Qwen3.5 models at Q4_K_M quantization, representing different points on the size–performance spectrum. The **headline result** is that **122B-class** weights can run locally on AMD Ryzen™ AI Max+ CPU with 128 GB UMA; most of the **hard numbers** below therefore cover 9B / 35B / 122B **end-to-end generation** (Ollama). The 9B and 35B models ran with 100% GPU offloading. The 122B model, at 76 GB, exceeds the 64 GB GPU-accessible memory allocation and uses a CPU/GPU mixed loading configuration (61% GPU, 39% CPU), which is automatically managed by Ollama.

Token Generation Speed

Model	Total Parameters	Active Parameters	Model Size	Generation Speed
Qwen3.5 9B	9B (dense)	9B	6.2 GB	29.84 tok/s
Qwen3.5 35B-A3B	35B (MoE)	3B	20.5 GB	42.04 tok/s
Qwen3.5 122B-A10B	122B (MoE)	10B	76 GB	8.59 tok/s (61% GPU / 39% CPU)

The 35B model achieves a higher generation speed than the 9B model due to its Mixture-of-Experts (MoE) architecture: despite having 35 billion total parameters, only 3 billion are active per token, resulting in faster inference.

The 122B model — at 76 GB — runs with CPU/GPU mixed loading. With a higher GPU memory allocation in the BIOS (depending on system vendor and BIOS support), more layers can be offloaded to the GPU for additional performance.

Figure 2 shows verbose generation statistics for the 35B MoE model; Figure 3 shows the 122B MoE model under mixed CPU/GPU loading.

```
prompt eval count:    30 token(s)
prompt eval duration: 194.153622ms
prompt eval rate:     154.52 tokens/s
eval count:           12970 token(s)
eval duration:        5m8.520282889s
eval rate:            42.04 tokens/s
```

Figure 2: `ollama run qwen3.5:35b --verbose` generation statistics on AMD Ryzen™ AI Max+ CPU. The 35B MoE model achieves 42.04 tok/s generation throughput and 154.52 tok/s prompt processing with 100% GPU offloading.

```
prompt eval count:    30 token(s)
prompt eval duration: 697.872736ms
prompt eval rate:     42.99 tokens/s
eval count:           8262 token(s)
eval duration:        16m1.916735289s
eval rate:            8.59 tokens/s
```

Figure 3: `ollama run qwen3.5:122b --verbose` generation statistics on AMD Ryzen™ AI Max+ CPU. The 122B MoE model achieves 8.59 tok/s generation throughput with CPU/GPU mixed loading (61% GPU / 39% CPU).

The exact hardware and software configuration used for these results is listed under the **Test environment disclosure** section at the end of this post.

Recommended Configurations

Use Case	Model	Expected Generation Speed	Notes
Fast interactive chat	Qwen3.5 35B-A3B	~42 tok/s	Strong speed-to-quality balance among the three checkpoints tested
Large-capacity workloads	Qwen3.5 122B-A10B	~8.6 tok/s	Largest checkpoint covered in this post
Lightweight tasks	Qwen3.5 9B	~30 tok/s	Smallest memory footprint of the three checkpoints tested

Memory Management Best Practices

With unified memory, the GPU and operating system share the same physical memory pool. For practical performance:

- **Monitor memory usage** with `rocm-smi` to verify GPU memory allocation during inference

- **Adjust context length** based on model size — larger models with long context windows consume more memory from the shared pool
- **Close memory-intensive desktop applications** when running the largest models to ensure sufficient memory is available for inference

Additional Resources

- [AMD ROCm Installation Guide](#)
- [Ollama](#)
- [AMD ROCm Developer Hub](#)

Summary

In this blog you explored what a 128 GB Unified Memory Architecture changes for local LLM inference on AMD Ryzen™ AI Max+ processor. You set up Ollama on Ubuntu with AMD ROCm™ software, then ran Qwen3.5 across 9B, 35B-A3B (MoE), and 122B-A10B (MoE) on a single system — fully GPU-offloaded for the smaller checkpoints, and CPU/GPU mixed for the 76 GB 122B-class model that Ollama transparently splits across the unified memory pool.

Key takeaways:

1. **128 GB unified memory enables 100B+ parameter models on a single system.** The Qwen3.5 122B model, at 76 GB, runs on AMD Ryzen™ AI Max+ processor with CPU/GPU mixed loading.
2. **Setup is straightforward.** Ollama provides a single-command installation; smaller checkpoints offload fully to the GPU, while the largest may use CPU/GPU mixed loading when they exceed the GPU-accessible memory slice.
3. **Performance is practical for interactive use.** The 35B model delivers ~42 tok/s generation speed, and the 122B model maintains ~8.6 tok/s for conversational workloads.

We are continuing to explore on-device AI on AMD Ryzen™ AI Max+ processors — including memory and BIOS tuning for longer context windows, throughput comparisons across additional model families, and end-to-end developer workflows on the AMD ROCm™ software platform. Stay tuned for upcoming posts from our team. In the meantime, download [Ollama](#) and try these Qwen3.5 checkpoints on your own AMD Ryzen™ AI Max+ processor-based system.

Test environment disclosure

All benchmarks were collected on AMD Ryzen™ AI Max+ 395 with 128 GB unified memory, 64 GB allocated to GPU, running Ubuntu 24.04 LTS, ROCm 7.2.1, and Ollama 0.20.x. Performance may vary based on system configuration, memory allocation, operating system, and software versions.

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